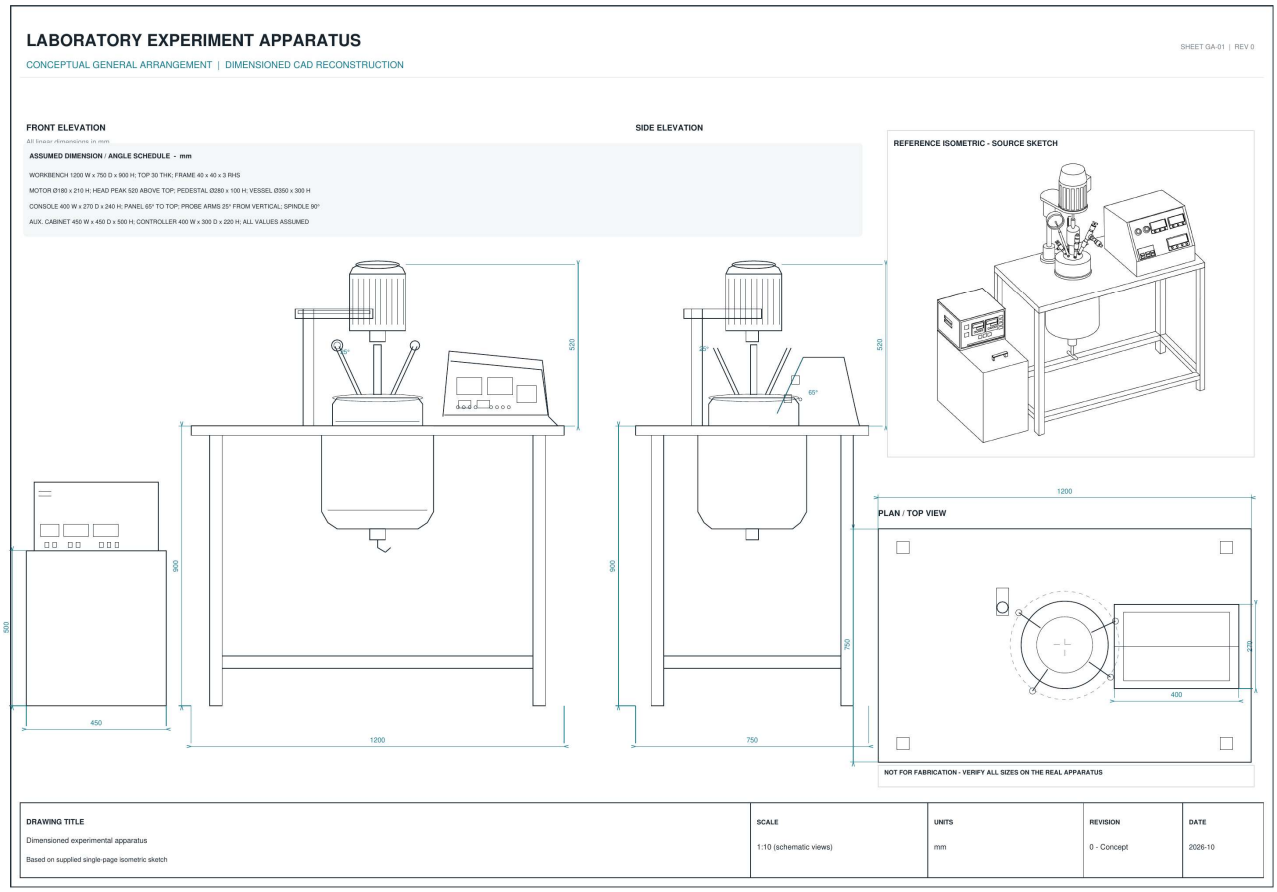
CAD RECONSTRUCTION & DIMENSIONING

Laboratory Experiment Apparatus | Conceptual General Arrangement



PROJECT REPORT | Drawing status: Conceptual | Units: mm | Revision: 0

Executive Summary

This project converts a supplied, unscaled isometric line sketch into a structured engineering drawing package for a laboratory experimental apparatus. The reconstruction interprets the visible arrangement as a bench-mounted motorised head, upper pedestal, suspended vessel, angled probe arms, inclined control console, steel-tube workbench and a separate auxiliary cabinet/controller. It establishes a consistent set of assumed dimensions and angles, then presents front, side and plan views with dimension lines, annotations, a title block and a source-sketch reference.

IMPORTANT DESIGN BASIS: The source sketch contains no scale, labels, apparatus name or verified dimensions. Dimensions and angles in this package are engineering assumptions for visualisation and CAD practice only. They must be measured and confirmed before fabrication, procurement, testing or safety approval.

Project at a Glance

Item	Description
Project type	2D CAD reconstruction and conceptual dimensioning
Source material	One-page isometric line-art sketch supplied by the project owner
Main outputs	Dimensioned drawing PDF, editable DXF, project report in DOCX/PDF
Core drafting skills	Orthographic views, dimensioning, angle callouts, drawing layers, assumptions register
Design status	Conceptual study - not a verified manufacturing drawing

1. Project Brief and Scope

1.1 Brief

The supplied diagram shows a laboratory arrangement with a central motor/shaft assembly positioned above a test vessel, an instrument console on the work surface, a framed bench, and a separate floor-level cabinet with a controller. The task was to retain these visible components while adding enough dimensional structure and angle information to make the drawing easier to interpret, review and continue editing in CAD software.

1.2 Objectives

- Translate the visual reference into front elevation, side elevation and plan views.
- Establish a coherent overall envelope and nominal component sizes using typical illustrative values.
- Dimension major lengths, heights, depths and diameters, and add angular callouts for the inclined panel and probe arms.
- Separate object geometry, dimensions, center/hidden features and annotation into CAD layers.
- Document key assumptions, project limits and the checks required before the drawing is used for physical work.

1.3 Scope Included

Included	Not included
Conceptual component layout and orthographic views	Identification of the actual experiment or machine model
Assumed dimensional schedule and angle schedule	Verified manufacturer specifications or as-built measurements
CAD layers, labels, dimension strings and title block	Electrical circuit, guarding, structural or process design
Editable 2D DXF plus print-ready PDF	Fabrication release, tolerances, GD&T or regulatory certification

1.4 Interpretation of the Sketch

The central assembly is represented as a motor-driven shaft/head above a circular pedestal and a suspended cylindrical vessel. The console is treated as an operator-facing display/control enclosure. The auxiliary floor cabinet is shown separately from the main workbench. These are geometric interpretations of the drawing, not confirmations of machine function.

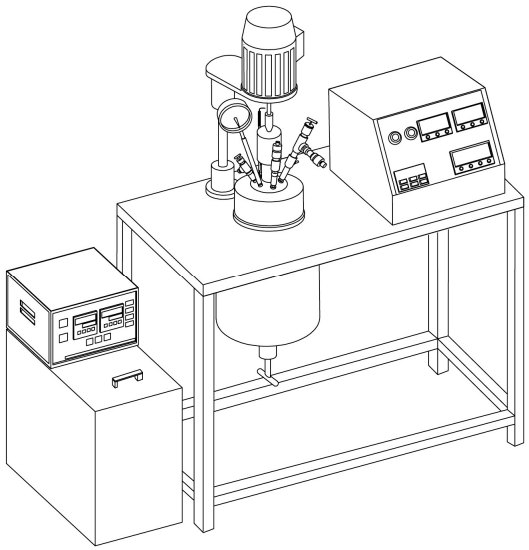


Figure 1. Source isometric sketch supplied for reconstruction.

2. Assumed Dimension Schedule

The values below establish a consistent visual model. They are nominal assumptions selected to create a plausible, legible layout from the unscaled image. They are not measurements of the original apparatus.

Component	Assumed dimensions (mm)	Notes
Main workbench	1200 W x 750 D x 900 H	Top surface to floor; overall main bench footprint
Worktop	1200 x 750 x 30 thick	Conceptual top plate/panel
Frame members	40 x 40 x 3 RHS	Assumed square hollow-section frame; material/grade unverified
Lower frame rail	40 x 40 section	Rail centreline approximately 140 mm above floor
Upper pedestal / base	Ø280 x 100 high	Mounted on worktop beneath the head/probe arrangement
Suspended vessel	Ø350 x 300 high	Nominal below-worktop cylindrical enclosure
Motor housing	Ø180 x 210 high	Nominal housing envelope; full head peak about 520 mm above worktop
Vertical support mast	Ø35 x approx. 380 exposed	Illustrative support member and bracket arrangement
Operator console	400 W x 270 D x 240 H	Mounted on worktop; inclined operator-facing panel
Auxiliary cabinet	450 W x 450 D x 500 H	Separate floor-level enclosure, shown beside bench
Auxiliary controller	400 W x 300 D x 220 H	Nominal enclosure placed on cabinet

2.1 Angle Schedule

Feature	Angle	Interpretation
Operator-facing console panel	65° to tabletop	Equivalent to 25° from vertical; conceptual side-profile angle
Probe/sensor arms	25° from vertical	Nominal layout angle; individual probe geometry not measured
Main spindle axis	90° to tabletop	Vertical motor/shaft axis in the reconstructed arrangement
Workbench legs	90° to tabletop	Vertical frame members
Workbench rails	0° to floor plane	Horizontal frame members

2.2 Dimensioning Convention

- All linear dimensions are expressed in millimetres (mm); diameters use the Ø symbol where shown.
- Overall envelope dimensions are prioritised; component detail is schematic and has no manufacturing tolerance.
- Angular values are shown as nominal design assumptions, not as angle measurements extracted from the source raster.

Before fabrication: measure overall footprint, bench height, vessel diameter/clearance, motor and spindle envelope, probe-arm positions, panel inclination, mounting holes, electrical clearances and all interfaces on the actual equipment.

3. CAD Methodology and Drawing Structure

Stage	Work performed
01 | Reference review	Inspected the supplied isometric drawing and listed the visible assemblies and their relative placement.
02 | Envelope definition	Selected a 1200 x 750 mm workbench and 900 mm working height as the baseline envelope; defined the adjacent cabinet and controller.
03 | Component geometry	Drafted the support frame, worktop, hanging vessel, pedestal, motor housing, mast, probe arms and control enclosures.
04 | Orthographic projection	Produced front, side and plan views so that overall width, depth and vertical clearances could be checked independently.
05 | Dimensioning and angles	Added overall dimensions, critical component diameters/heights, a 65° console panel angle and 25° nominal probe-arm angle.
06 | File/package preparation	Created a print-ready drawing PDF and an editable DXF model-space drawing, then documented assumptions and limitations.

3.1 Proposed CAD Layer Convention

Layer	Purpose	Drafting treatment
OBJECT	Primary visible component outlines	Continuous; main lineweight
THIN	Secondary edges, component details	Continuous; light lineweight
DIMENSIONS	Dimension lines, extension lines, leaders and arrows	Distinct cyan/blue colour in CAD file
CENTER	Axes and centre marks	Center line type where available
HIDDEN	Projected vessel boundary and obscured features	Dashed line type where available
TEXT	View labels, notes and drawing title	Readable annotation height

3.2 Drawing Outputs

File	Use
apparatus_dimensioned_drawing.pdf	A3 landscape drawing sheet for review and print; contains three orthographic views, dimensions, angle notes and source isometric inset.
apparatus_dimensioned_drawing.dxf	Editable CAD entities arranged in model space at full nominal size; use AutoCAD/DraftSight/LibreCAD to inspect or revise geometry.
apparatus_cad_project_report.docx	Editable narrative report with scope, assumptions, methodology, schedules and CV-ready summary.
apparatus_cad_project_report.pdf	Fixed-layout copy of the report for sharing or portfolio submission.

3.3 Quality Checks Applied

- Cross-checked the overall dimensions against the declared assumed envelope: 1200 mm bench width, 750 mm depth, 900 mm worktop height and approximately 520 mm equipment height above the worktop.
- Kept the same assumed critical dimensions in the drawing annotations and report tables.
- Marked the source view as a reference and distinguished conceptual geometry from measured geometry.
- Reviewed rendered output for page fit, legibility and visible clipping before delivery.

Conclusion

This package demonstrates a workflow from a visual reference to an editable, dimensioned concept drawing. Present it as a CAD reconstruction exercise, not as a tested or fabrication-ready machine design.